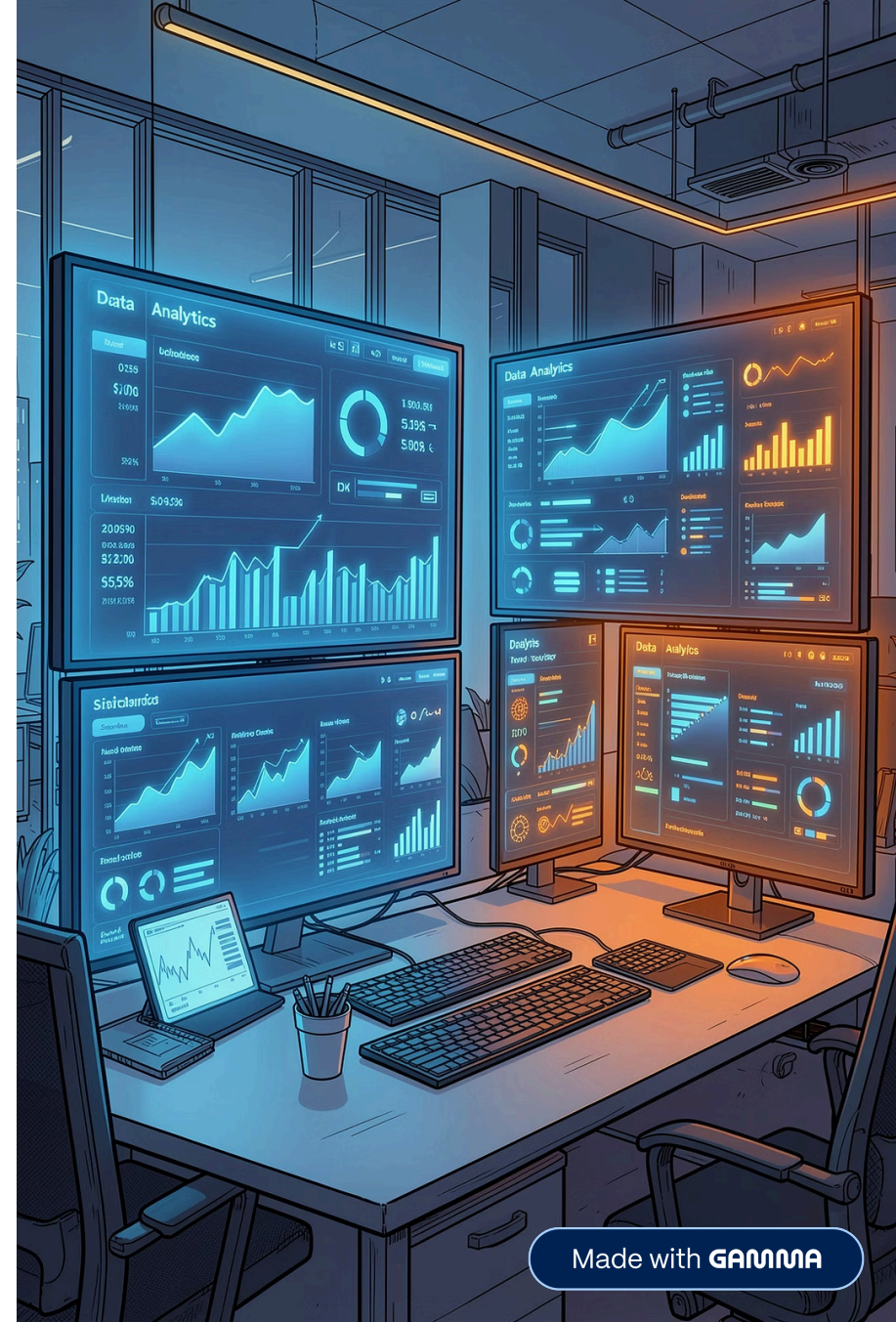


Customer Behavior Data Analysis

End-to-end data analysis pipeline transforming **3,900 customer records** across **19 attributes** into actionable business intelligence — powered by Python, PostgreSQL, and Power BI.

JUNE 2026

COMPLETE PROJECT



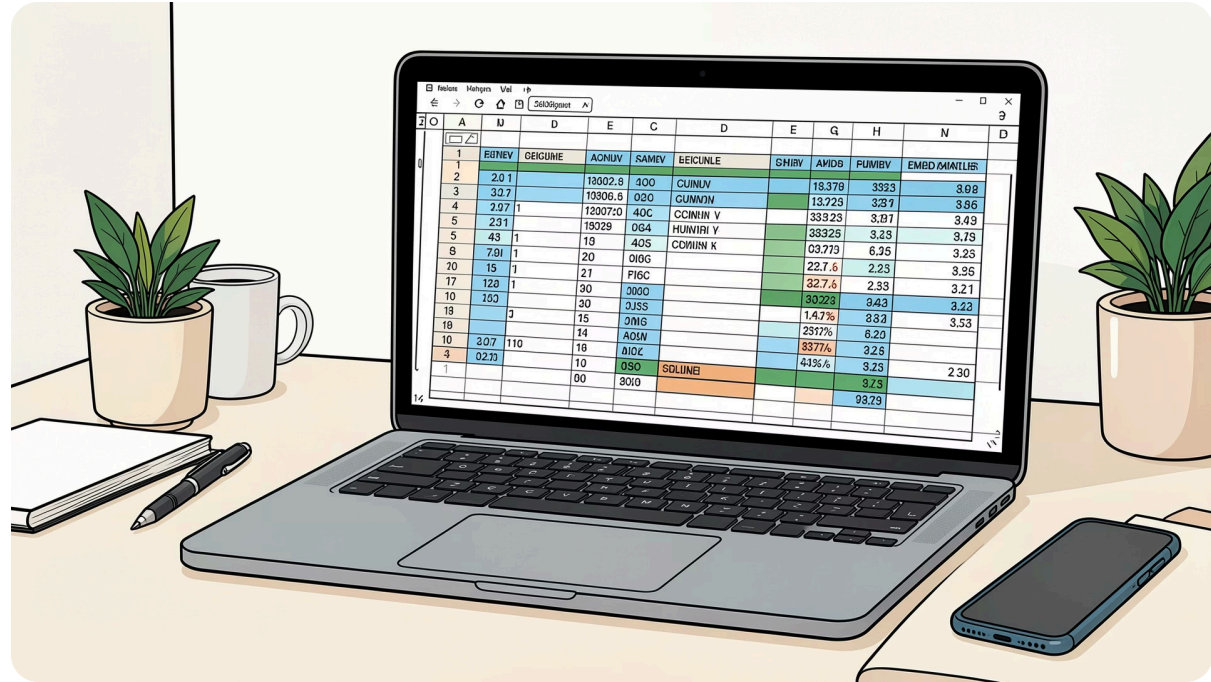
Made with GAMMA

Dataset Overview

3,900 Records · 18
Attributes + 2 Engineered

E-commerce transaction data
covering customer demographics,
purchase details, product
attributes, and behavioral metrics.

- Age range: 18–65 years
- 4 categories: Clothing, Accessories, Footwear, Outerwear
- Review Rating: 0–5 scale (37 missing = 0.9%)
- Engineered features: Age Group, Purchase Freq Days



📄 **Data Quality:** 99.1% complete · All categorical variables encoded · No data type inconsistencies

Steps 1–2: Loading & Data Quality

✓ Data Loaded

3,900 records, 18 columns.

Customer ID 1–3900, Age 18–65.

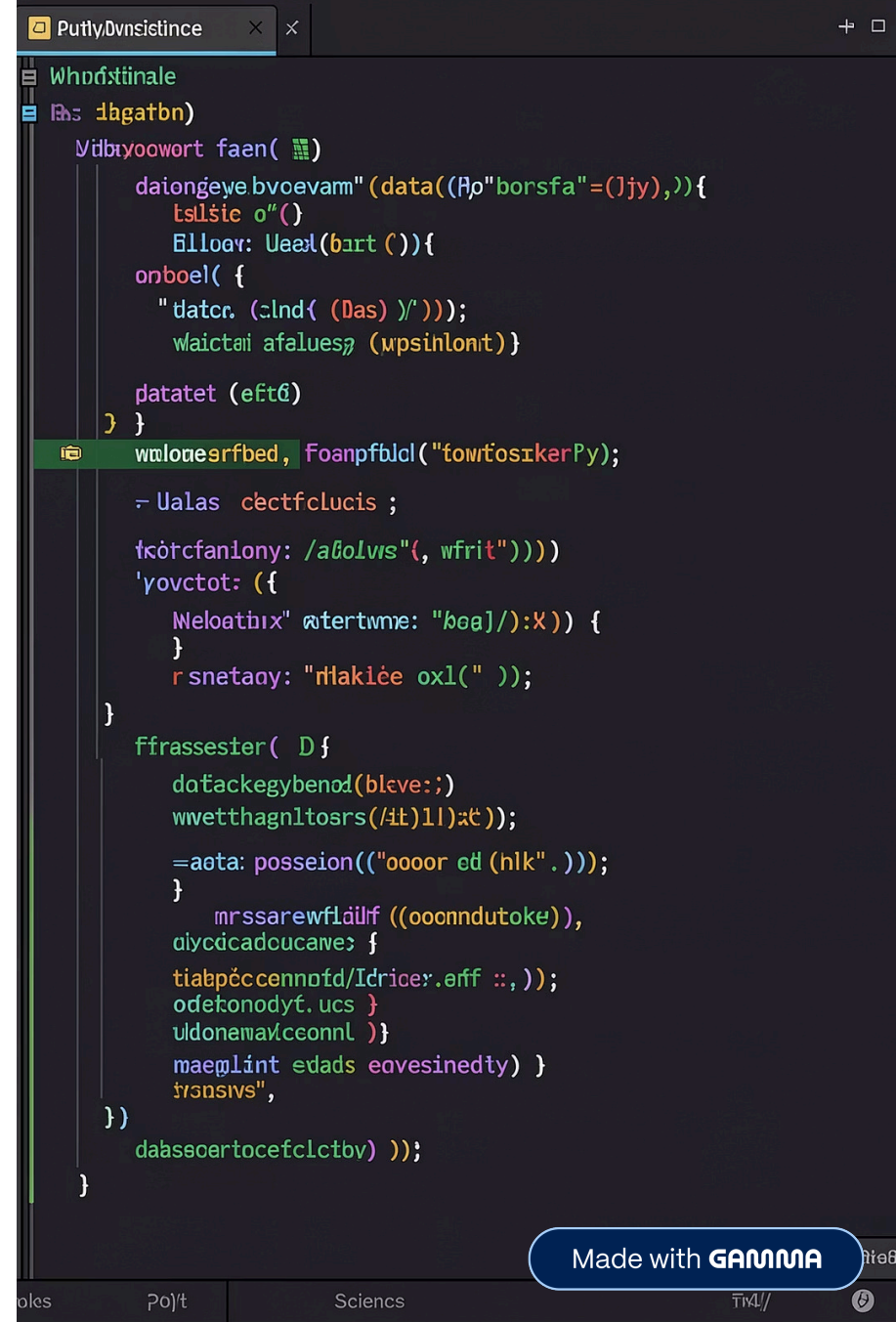
Memory: 608 KB.

✓ Redundant Column Removed

'Promo Code Used' 100% correlated with 'Discount Applied.' Dropped – 17 columns retained.

✓ Missing Values Assessed

Review Rating: 37 missing (0.95%) – below 1% threshold. Records retained as-is.



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Made with GAMMA

Step 3: Feature Engineering

Age Group Segmentation

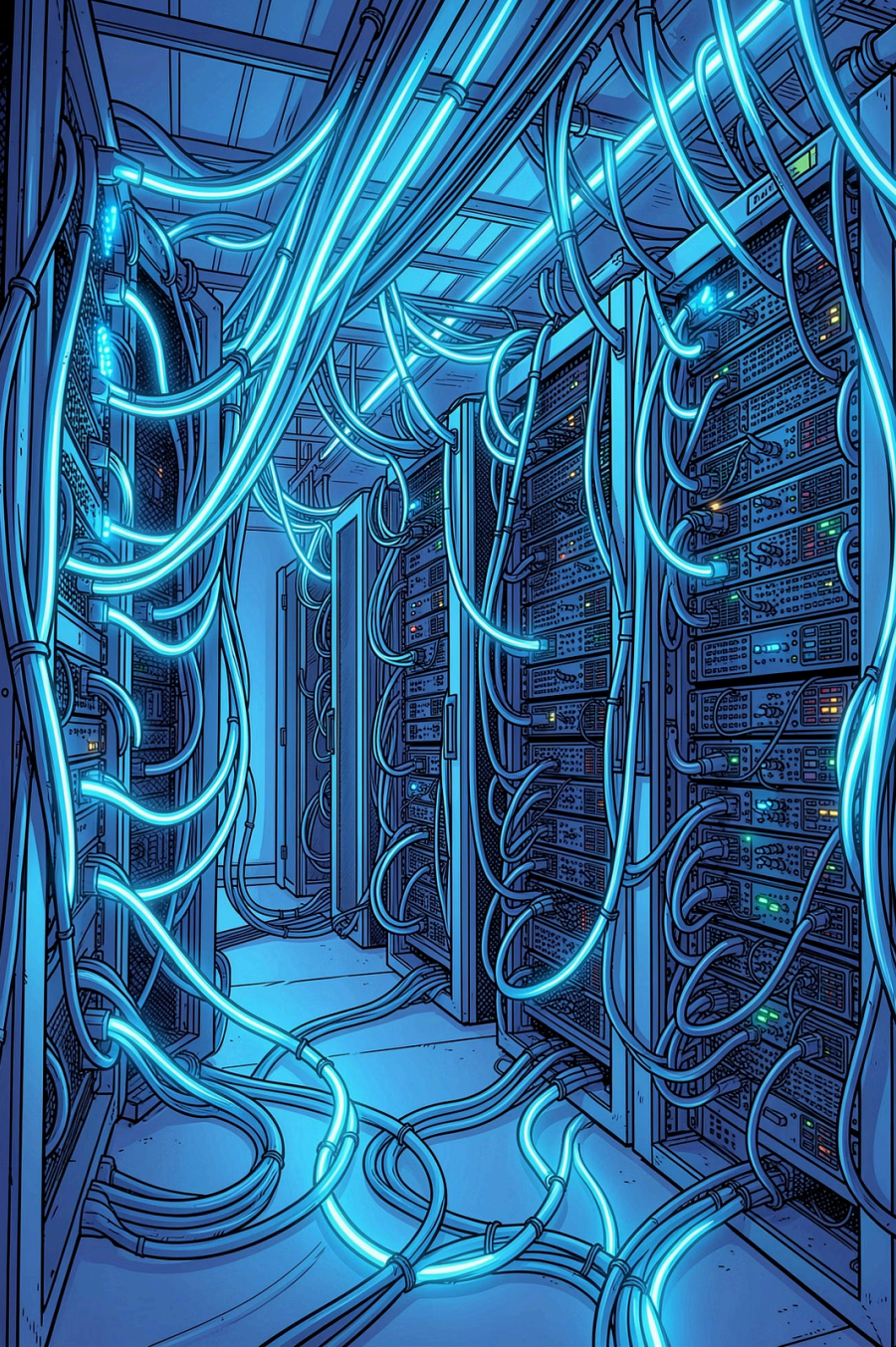
Continuous age mapped to 4 demographic segments for targeted analysis:

- **Young Adult** (18–24): 715 customers — 18%
- **Middle-aged** (25–39): 1,189 customers — 30%
- **Adult** (40–54): 1,232 customers — 32%
- **Senior** (55–65): 764 customers — 20%

Purchase Frequency Conversion

Categorical frequency mapped to numeric days for quantitative modeling:

- Weekly → 7 days
- Bi-Weekly / Fortnightly → 14 days
- Monthly → 30 days
- Quarterly → 90 days
- Annually → 365 days



Step 4: PostgreSQL Integration

1

Connect

SQLAlchemy ORM to localhost:5432 · database: *customersdatabase*

2

Load

```
df.to_sql("customer", if_exists="replace")
```

3

Deploy

3,900 records · 19 columns · Power BI connection ready

Power BI Dashboard & KPIs



Key Performance Indicators

4 interactive filters: Subscription Status, Gender, Category, Shipping Type

3.9K Customers

Total customer base analyzed

\$59.76 Avg Purchase

Moderate purchasing power –
upselling opportunity

3.75 Avg Rating

Below 4.2 industry benchmark
– quality gap

Dashboard Visualizations

Subscription Status

Donut chart: **27% subscribers** (1,053) vs **73% non-subscribers** (2,847) — massive conversion opportunity.

Revenue by Category

Clothing leads at 40% (~\$2,000), followed by Accessories 26%, Footwear 18%, Outerwear 14%. Category imbalance creates risk.

Revenue by Age Group

Middle-aged (25–39) generate **highest revenue** — premium segment despite not being the largest. Young Adults show lowest revenue but untapped potential.

Sales by Age Group

Middle-aged also lead in transaction frequency and buy higher-value items — validated target for premium positioning.

6 Strategic Insights

1

Subscription Growth

27% rate – convert 10% of non-subscribers = **\$500K–\$750K** recurring revenue

2

Category Imbalance

Clothing 40% vs Outerwear 14% – rebalance to 25–30% per category

3

Age-Based Spending

Middle-aged (25–39) highest spenders – personalize for **20–25% conversion lift**

4

Satisfaction Gap

3.75 vs 4.2 benchmark – close gap for **15–20% conversion increase**

5

Seasonal Patterns

Plan inventory 2–3 months ahead to reduce stock-outs and excess

6

Engagement Variability

Weekly to Annual frequency range – deploy re-engagement and loyalty programs

Strategic Recommendations & ROI

① Subscription Program

Budget: **\$60K** → \$85.5K annual revenue · **142% ROI** · Multi-year: \$255K+

② Category Rebalancing

Invest: **\$30K** → Outerwear 14% → 20% · **1,000% ROI** via SKU expansion & bundling

③ Demographic Marketing

Budget: **\$40K** → age-specific campaigns · **20-25% conversion lift** = \$200K+ impact

④ Quality Initiative

Budget: **\$25K** → rating 3.75 → 4.2 · **300%+ ROI** via retention & conversion gains

Project Summary

Key Accomplishments

- Loaded, cleaned & engineered 3,900 records
- PostgreSQL production database deployed
- Power BI dashboard with 5+ visualizations
- 6 strategic insights · 4 ROI-backed recommendations

Technology Stack

Python 3.11 + Pandas · Jupyter · PostgreSQL + SQLAlchemy ·
Power BI

Total Revenue Opportunity

\$1.2M+

- Subscription growth: \$500K–\$750K
- Category optimization: \$200K+
- Targeted marketing: \$200K+
- Quality improvements: \$150K+

■ **COMPLETE — Ready for Production Deployment**

📌 **Next Steps:** Deploy dashboard · Launch subscription campaign · Initiate quality task force · Begin category rebalancing · Implement demographic targeting